MSE 311 Thermodynamics of materials-Homework 14

Name: *Student ID:*

Please hand in this homework before class on December 22th.

1. When a weight was hung on a pure Cu wire of radius r = 0.1 mm and held at a high temperature, the balanced load of the weight was $m = 5.5\times10^{-5}$ kg, which caused neither expansion nor shrinkage of the wire. Estimate the surface energy of pure Cu from this experimental value.

$$\nu = \frac{\frac{\Delta r}{r}}{\frac{\Delta l}{l}} = \frac{1}{2} \Rightarrow \Delta r = \frac{\Delta l}{2l} r$$

$$mg\Delta l = \gamma \Delta A = \gamma\left[2\pi(r-\Delta r)(l+\Delta l) - 2\pi r l\right]$$

$$= \gamma 2\pi(r\Delta l - l\Delta r - \Delta r \Delta l) \approx \gamma 2\pi\left(r\Delta l - \frac{\Delta l}{2} r\right) = \gamma\pi r \Delta l$$

$$mg = \gamma \pi r$$

$$\gamma = \frac{mg}{\pi r} = \frac{5.5\times10^{-5}\times9.8}{3.14\times0.1\times10^{-3}} = 1.7 J/m^2$$

2. Steel can be hardened and made more corrosion resistant by heat treating in a nitrogen atmosphere to diffuse nitrogen into the surface laver of the steel. This process can be roughly modeled using error function solution as following.

$$C(y,t) = A + B\text{erf}\left(\frac{y}{\sqrt{4Dt}}\right)$$

Assume that a plate of steel is heat treated in a nitrogen environment at 500°C, where the solubility of nitrogen in steel is $C_N^* \approx 0.5$ mol % and the initial background concentration of nitrogen in the steel is zero ($C_N^0 = 0$).

(a) Provide the boundary condition and initial condition for this problem.

(b) Based on your boundary and initial conditions, provide the *A* and *B* of error function solution in this problem.

(c) Assuming the diffusivity of nitrogen in steel is $D_{N/steel} = 10^{-9}$ cm²/s at the heat treatment temperature, determine how long it will take for the nitrogen concentration 10 μm deep into the steel to reach 70% of its value on the surface (i.e., 70% of C_N^*)?

(d) How long would it take the nitrogen concentration 20 μm deep into the steel to reach 70% of its value on the surface?

(a) Boundary condition: C(0,t)=0.5 mol%
Initial condition: y>0, C(y,0)=0

(b) $\begin{cases} C(0,t) = A + B\mathrm{erf}\,0 = A = 0.5 \\ C(y,0) = A + B\mathrm{erf}\infty = A + B = 0 \end{cases} \Rightarrow \begin{cases} A = C_N^* = 0.5 \\ B = -C_N^* = -0.5 \end{cases}$

(c) $0.7C_N^* = C_N^* - C_N^*\mathrm{erf}\left(\dfrac{10\times10^{-6}}{\sqrt{4\times10^{-9}\times10^{-4}t}}\right) \Rightarrow t = 3200s = 53\min$

(d)
$$\frac{10\times10^{-6}}{\sqrt{4\times10^{-9}\times10^{-4}t_1}} = \frac{20\times10^{-6}}{\sqrt{4\times10^{-9}\times10^{-4}t}}$$
$$t = \left(\frac{20}{10}\right)^2 t_1 = 4t_1 = 212\min = 3.53h$$

3. Estimate the following parameters in solidification of pure water by the values in following Table 1. (a) the change of Gibbs energy of melting (ΔG_V), (b) critical nucleation size (r^*) and the change of Gibbs energy to for a ordering crystalline cluster in a diameter of r^* (ΔG_r^*), (c) frequency of atomic transition (ν_c) and homogeneous nucleation rate (I). The diffusion coefficient of the liquid phase is presumed as $D^L \approx 10^{-9}$ m^2/s. In the case of ice, the volume of a molecule is 3×10^{-29} m^3.

Table 1 Calculated values for parameters of water regarding solidification

Substance	**Melting point (T_m), K**	**Latent heat of fusion (L_V), kJ/mol**	**Molar volume (V), m^3/mol**	**Interface energy (γ), J/m^2**	**Maximum degree of supercooling (ΔT_{max}), K**
H_2O	273	6.0	18×10^{-6}	0.025	40

(a) $\Delta G_V = \dfrac{L_V \Delta T_{\max}}{T_m V} = \dfrac{6000\times40}{273\times18\times10^{-6}} = 4.884\times10^7\,J/m^3$

(b)
$$r^* = \frac{2\gamma}{\Delta G_V} = \frac{2\times0.025}{4.884\times10^7} = 1.024\times10^{-9}m = 1.024nm$$
$$\Delta G_r^* = \frac{4\pi}{3}r^{*2}\gamma = \frac{4\times3.14}{3}\times\left(1.024\times10^{-9}\right)^2\times0.025 = 1.098\times10^{-19}J$$

(c)

$$r_a = \sqrt[3]{\frac{3V}{4\pi}} = \sqrt[3]{\frac{3\times3\times10^{-29}}{4\times3.14}} = 1.928\times10^{-10}m = 0.1928nm$$

$$v_c = \frac{\pi r^{*2}}{4r_a^4}D^L = \frac{3.14\times\left(1.024\times10^{-9}\right)^2}{4\times\left(1.928\times10^{-10}\right)^4}\times10^{-9} = 5.96\times10^{11}s^{-1}$$

$$N_c = N\exp\left[-\frac{\Delta G_r^*}{k_B\left(T_m-\Delta T_{\max}\right)}\right] = 6.022\times10^{23}\times\exp\left[-\frac{1.098\times10^{-19}}{1.38\times10^{-23}\times\left(273-40\right)}\right] = 8.9\times10^8 mol^{-1}$$

$$I = \frac{N_c}{V}v_c = \frac{8.9\times10^8}{18\times10^{-6}}\times5.96\times10^{11} = 2.947\times10^{25}m^{-3}\cdot s^{-3}$$

4. The nucleation process of pure metal solidification usually requires a large degree of supercooling, but the concept of "superheat" is rarely mentioned when pure metal is melted. Why?

5. Figure 1 gives a set of rate of transformation curves for the crystallization of a glass. Using this information, calculate and plot a TTT (time–temperature–transformation) diagram for this crystallization that includes curves for 10, 50, and 90% transformed with temperature on the vertical axis versus log (time) on the horizontal axis.

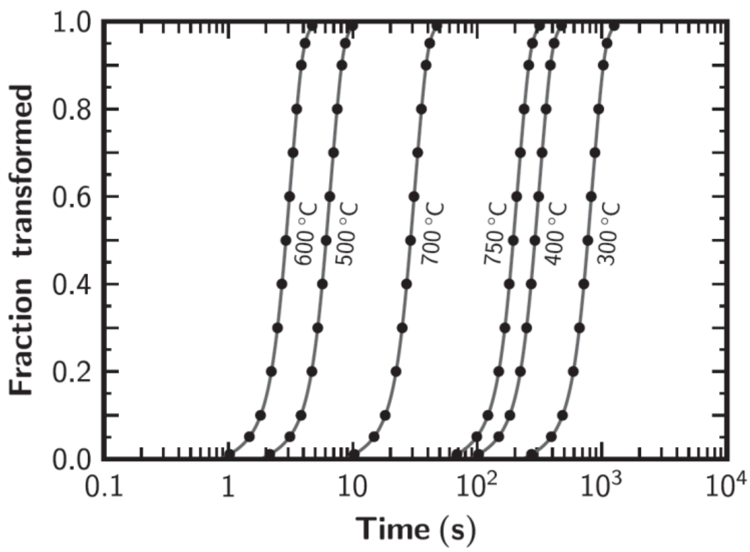

Figure 1 Crystallization of a glass as a function of time for various transformation temperatures.

6. Al_2O_3, which melts at 2324 K, and Cr_2O_3, which melts at 2538 K form complete ranges of solid and liquid solutions. Assuming that $\Delta S^{\circ}_{m,Cr_2O_3} = \Delta S^{\circ}_{m,Al_2O_3}$, and that the solid and liquid solutions in the system Al_2O_3-Cr_2O_3 behave ideally. $\Delta H^{\circ}_{m(Al_2O_3)} = 107.5 kJ/kg$. When an alloy of $X_{Al_2O_3} = 0.5$ is heated, the temperature at which equilibrium melting is beginning and complete is 2420K and 2444K, respectively. Calculate

(1) $\Delta G^{\circ}_{m(Al_2O_3)}$ and $\Delta G^{\circ}_{m(Cr_2O_3)}$ in terms of temperature (T).

(2) The composition of the melt which first forms.

$$X_{A(l)} = \frac{\left[1-\exp\left(-\Delta G^{\circ}_{m(B)}/RT\right)\right]\exp\left(-\Delta G^{\circ}_{m(A)}/RT\right)}{\exp\left(-\Delta G^{\circ}_{m(A)}/RT\right)-\exp\left(-\Delta G^{\circ}_{m(B)}/RT\right)}$$

$$X_{A(s)} = \frac{1-\exp\left(-\Delta G^{\circ}_{m(B)}/RT\right)}{\exp\left(-\Delta G^{\circ}_{m(A)}/RT\right)-\exp\left(-\Delta G^{\circ}_{m(B)}/RT\right)}$$

$$\Delta G^{\circ}_{m(i)} = \Delta H^{\circ}_{m(i)}\left(\frac{T_{m(i)}-T}{T_{m(i)}}\right)$$

$$\Delta G^{\circ}_{m(Al_2O_3)} = \Delta H^{\circ}_{m(Al_2O_3)}\left(\frac{T_{m(Al_2O_3)}-T}{T_{m(Al_2O_3)}}\right) = 107500 - \frac{107500}{2324}T = 107500 - 46.26T$$

(1) $\Delta S^{\circ}_{m,Al_2O_3} = \Delta S^{\circ}_{m,Cr_2O_3} \Rightarrow \dfrac{\Delta H^{\circ}_{m(Al_2O_3)}}{T_{m(Al_2O_3)}} = \dfrac{\Delta H^{\circ}_{m(Cr_2O_3)}}{T_{m(Cr_2O_3)}}$

$$\Delta G^{\circ}_{m(Cr_2O_3)} = \Delta H^{\circ}_{m(Cr_2O_3)} - \frac{\Delta H^{\circ}_{m(Cr_2O_3)}}{T_{m(Cr_2O_3)}}T = \frac{\Delta H^{\circ}_{m(Al_2O_3)}}{T_{m(Al_2O_3)}}T_{m(Cr_2O_3)} - \frac{\Delta H^{\circ}_{m(Al_2O_3)}}{T_{m(Al_2O_3)}}T = 117399 - 46.26T$$

$$X_{Al_2O_3(l)} = \frac{\left[1-\exp\left(-\Delta G^{\circ}_{m(Cr_2O_3)}/RT\right)\right]\exp\left(-\Delta G^{\circ}_{m(Al_2O_3)}/RT\right)}{\exp\left(-\Delta G^{\circ}_{m(Al_2O_3)}/RT\right)-\exp\left(-\Delta G^{\circ}_{m(Cr_2O_3)}/RT\right)}$$

$$= \frac{\left[1-\exp\left(-\frac{14120}{2420}+5.56\right)\right]\exp\left(-\frac{12930}{2420}+5.56\right)}{\exp\left(-\frac{12930}{2420}+5.56\right)-\exp\left(-\frac{14120}{2420}+5.56\right)} = 0.618$$